

Source Integration

- NCERT Class 12 Introductory Macroeconomics, Chapter 2: National Income Accounting

Core Factual & Conceptual Architecture

1. Foundational Classifications

- Final Goods: Commodities and services purchased for ultimate consumption or investment that will undergo no further production or resale within the domestic economic boundary. Classification is determined strictly by economic use rather than physical properties.
 - Consumption Goods: Goods consumed directly by households to satisfy immediate wants (such as food, apparel, recreation).
 - Capital Goods: Durable producer goods (machinery, factory equipment, physical tools) that aid continuous production across multiple cycles without being consumed themselves, depreciating over time.
 - Consumer Durables: Final consumer products that yield utility over a multi-year lifespan (automobiles, domestic appliances).
- Intermediate Goods: Raw materials or semi-finished items fully utilized or transformed during production within the accounting year. Excluded from national income accounting to prevent double counting.
- Stock vs. Flow Framework:
 - Stock: Economic magnitude measured at a specific single point in time (such as total capital stock, national wealth, year-end inventory).
 - Flow: Economic magnitude generated and measured across a defined period of time (such as monthly wages, annual GDP, annual investment).

2. Capital Accumulation & Depreciation

- Gross Investment: Total capital expenditure incurred across the economy within a year to acquire physical productive capital.
- Depreciation: Regular consumption of fixed capital due to standard wear and tear and anticipated obsolescence during production.
- Net Investment: The genuine net addition made to an economy's productive capital stock: $\text{Net Investment} = \text{Gross Investment} - \text{Depreciation}$.
- Inventory Changes: Finished and work-in-progress stock held by enterprises constitutes a stock variable, but the year-over-year change in inventory represents an investment flow.

3. The Circular Flow & Three Measurement Methods

- Two-Sector Circular Flow: Real flows of factors of production from households to firms alongside reverse factor income flows, matched by real output flows from firms to households matched by consumption outlays.
- The Three Equivalent Approaches:
 1. Product / Value-Added Method: Summation of gross value added across all productive entities ($\text{Value Added} = \text{Gross Value of Output} - \text{Intermediate Consumption}$) to eradicate double counting.
 2. Income Method: Summation of all factor earnings disbursed across productive entities ($\text{Compensation of Employees} + \text{Operating Surplus} [\text{Rent} + \text{Interest} + \text{Profit}] + \text{Mixed Income}$).
 3. Expenditure Method: Summation of final expenditures incurred on domestic output: $\text{GDP at Market Price} = C + I + G + (X - M)$, where C is private final consumption, I is investment, G is government consumption, and (X - M) represents net exports.

4. The National Accounting Hierarchy

- GDP at Market Price (GDP_MP): Market value of all final goods and services produced within the domestic boundary over an accounting period.
- GDP at Factor Cost (GDP_FC): $\text{GDP_MP} - \text{Net Indirect Taxes (NIT)}$, stripping out indirect taxes and adding back production subsidies.
- NDP at Market Price (NDP_MP): $\text{GDP_MP} - \text{Depreciation}$.
- NDP at Factor Cost (NDP_FC): Domestic factor income, derived as $\text{NDP_MP} - \text{NIT}$.
- GNP at Market Price (GNP_MP): $\text{GDP_MP} + \text{Net Factor Income from Abroad (NFIA)}$.
- NNP at Factor Cost (NNP_FC): National Income (NI), derived as $\text{NNP_MP} - \text{NIT}$ or $\text{NDP_FC} + \text{NFIA}$.
- Personal Income (PI): $\text{NI} - \text{Undistributed Corporate Profits} - \text{Net Interest Paid by Households} - \text{Corporate Taxes} + \text{Transfer Payments}$.
- Personal Disposable Income (PDI): $\text{PI} - \text{Personal Direct Taxes} - \text{Non-Tax Administrative Fines}$.

5. Price Deflators & Inflation Metrics

- Nominal GDP: Domestic production valued using current market prices.
- Real GDP: Domestic production valued using constant base-year prices, isolating genuine volume changes from price distortion.
- GDP Deflator: The ratio of Nominal GDP to Real GDP: $(\text{Nominal GDP} / \text{Real GDP}) * 100$, tracking comprehensive domestic price changes across all produced commodities.
- CPI vs. WPI: Consumer Price Index measures the cost of a representative consumer consumption basket including services, whereas the Wholesale Price Index captures wholesale bulk commodity transactions and omits services.

6. Limitations of GDP as a Measure of Social Welfare

- Inequality of Distribution: Aggregate GDP growth does not reflect distributional skew across population quintiles.
- Non-Monetized Economy: Omission of subsistence agriculture, domestic care work, and informal barter distorts accounting in developing economies.
- Negative Externalities: Environmental degradation, pollution, and resource depletion are not deducted, leading standard GDP to overstate actual societal welfare.